

RESTful Booker API - Test Plan

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Objective

The objective of this test plan is to verify the RESTful Booker API functionality strictly based on the provided PRD/API documentation.

The testing scope includes authentication APIs, booking CRUD operations, request and response validation, authorization checks, and API behavior validation. API automation using REST Assured framework is included as mentioned in the requirements.

No functionality or behavior outside the PRD has been assumed.

URL - <https://restful-booker.herokuapp.com/apidoc/index.html>

Scope

1. Functional Testing:

- Verify API endpoints based on the API documentation.
- Validate booking creation, update, retrieval, and deletion scenarios.
- Validate authentication and authorization for protected APIs.

2. Data Validation Testing:

- Validate mandatory fields and request payload validations.
- Validate response payload accuracy.
- Validate handling of invalid request data.

3. Error Handling Testing:

- Verify behavior for malformed or invalid requests.
- Validate error handling based on available API documentation.
- Exact error schema is not defined in PRD and remains TBD.

4. Performance Testing:

- Validate API behavior under execution load.
- Validate concurrent request handling.
- SLA thresholds are not explicitly defined in PRD.

5. Security Testing:

- Validate authentication and authorization behavior.
- Validate secure transmission practices.
- Perform SQL Injection and XSS validation as referenced in PRD.

6. Integration Testing:

- Verify interactions between API endpoints.

7. Compatibility Testing:

- Validate API execution on supported platforms/tools referenced in sample.

8. Documentation Review:

- Validate alignment between API documentation and API behavior.

9. Load Testing:

- Validate API behavior under concurrent load conditions.

10. Regression Testing:

- Validate CRUD operations after fixes or updates.

11. Edge Case Testing:

- Validate boundary and edge case request scenarios.

12. Concurrency Testing:

- Validate simultaneous CRUD operation handling.

13. Ad Hoc Testing:

- Exploratory validation based on observed API behavior.

14. Continuous Integration and Deployment (CI/CD) Testing:

- Validate automation execution within CI/CD flow where applicable.

15. Rate Limiting Testing:

- Rate limiting thresholds are not explicitly defined in PRD and remain TBD.

Inclusions

Create (POST) Operations:

- Validate booking creation using valid payloads.
- Validate mandatory field handling.
- Validate invalid request handling.

Read (GET) Operations:

- Validate booking retrieval by booking ID.
- Validate retrieval using available filters/query parameters.
- Validate behavior for invalid booking IDs.

Update (PUT/PATCH) Operations:

- Validate complete update functionality.
- Validate partial update functionality.
- Validate authorization validation for update operations.

Delete (DELETE) Operations:

- Validate booking deletion functionality.
- Validate response after successful deletion.

Boundary Testing:

- Validate minimum and maximum field values.

Concurrency Testing:

- Validate simultaneous CRUD execution handling.

Data Validation:

- Validate invalid characters, invalid data types, and mandatory fields.

Authentication and Authorization:

- Validate authenticated and unauthenticated API access.

Error Handling:

- Validate behavior for malformed requests.

Security Testing:

- Validate SQL Injection and XSS scenarios referenced in PRD.

Performance Testing:

- Validate API response behavior under load.

Integration Testing:

- Validate interaction between related APIs.

Regression Testing:

- Validate existing functionality after updates.

Compatibility Testing:

- Validate execution on supported platforms/tools.

Continuous Integration and Deployment (CI/CD) Testing:

- Validate automation execution in CI/CD flow.

Rate Limiting Testing:

- Rate limit expectations remain TBD as not explicitly defined in PRD.

Test Environments

QA Environment:

<https://restful-booker.herokuapp.com/apidoc/index.html>

Pre-Prod Environment:

<https://restful-booker.herokuapp.com/apidoc/index.html>

Supported platforms/tools referenced in sample:

- Windows 10 – Chrome, Firefox, Edge
- Mac OS – Safari
- Android Mobile OS – Chrome
- iPhone Mobile OS – Safari

Additional Environment Details:

- Authentication handling approach – TBD
- Rate limiting thresholds – TBD
- Test data reset strategy – TBD

Defect Reporting Procedure

- Defects will be logged for deviations from documented API behavior.
- Detailed reproduction steps, request payloads, response payloads, and logs will be attached.
- Defect management tooling remains TBD as not explicitly defined in PRD.
- Severity and priority assignment process remains TBD.

Defect Process POC:

Frontend – TBD

Backend – TBD

DevOps – TBD

Tools:

- TBD / Not provided in PRD

Test Strategy

Step 1:

Create API test scenarios and API test cases based on the documented requirements.

Testing techniques:

- Equivalence Class Partitioning
- Boundary Value Analysis
- Decision Table Testing
- Error Guessing
- Exploratory Testing

Step 2:

Testing procedure:

- Execute smoke validation before detailed testing.
- Reject unstable builds/environments if smoke validation fails.
- Execute detailed functional and regression validation after stable environment confirmation.
- Multiple resources may execute APIs simultaneously where concurrency validation is required.

Testing Types:

- Smoke Testing
- Sanity Testing
- Functional Testing
- Regression Testing
- Retesting
- API Automation Testing
- Security Validation
- Load and Concurrency Validation

Automation Scope:

- REST Assured framework
- API request/response validation
- Regression suite execution
- CI/CD integration validation

Step 3:

Best Practices:

- Context Driven Testing
- Shift Left Testing
- Exploratory Testing
- End-to-End Flow Validation

Test Schedule

Task Time Duration

- Creating Test Plan
- API Test Scenario Creation
- API Test Case Creation

- Automation Script Development
- Test Execution
- Defect Reporting
- Summary Report Submission

Schedule / Sprint timelines – TBD as not explicitly defined in PRD.

Test Deliverables

- Test Plan Document
- API Test Scenarios
- API Test Cases
- Automation Scripts
- Defect Reports
- Test Summary Report

Entry and Exit Criteria

Requirement Analysis

Entry Criteria:

- PRD/API documentation available.

Exit Criteria:

- Requirements reviewed and understood.
- Clarifications identified and documented.

Test Execution

Entry Criteria:

- API endpoints accessible.
- Stable environment available.
- Test cases prepared.

Exit Criteria:

- Planned test execution completed.
- Defects logged and tracked.

Test Closure

Entry Criteria:

- Test execution completed.
- Reports prepared.

Exit Criteria:

- Test Summary Report completed.

Tools

The following tools/frameworks are included based on requirements:

- Postman
- REST Assured
- Java
- Maven
- Git

CI/CD tooling remains TBD because PRD does not define implementation details.

Risks and Mitigations

Risk: Missing requirement details in PRD

Mitigation: Requirement clarification before execution.

Risk: Environment instability

Mitigation: Validate environment before execution.

Risk: Undefined SLA/performance thresholds

Mitigation: Record observations without assuming expected values.

Risk: Limited testing timelines

Mitigation: Prioritize critical API flows and regression coverage.

Approvals

The following deliverables may require stakeholder approval:

- Test Plan
- API Test Scenarios
- API Test Cases
- Automation Coverage
- Reports

Stakeholder names and approval flow are not explicitly defined in PRD and remain TBD.